

Research paper

Multi-modal signal integration for enhanced sleep stage classification: Leveraging EOG and 2-channel EEG data with advanced feature extraction

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ABSTRACT

This paper introduces an innovative approach to sleep stage classification, leveraging a multi-modal signal integration framework encompassing Electrooculography (EOG) and two-channel electroencephalography (EEG) data. We explore the utility of various feature extraction techniques, including Short-Time Fourier Transform (STFT), Wavelet Transform, and raw signal processing, alongside the utilization of neural networks as feature extractors. This unique combination allows us to harness the benefits of traditional feature extraction methods while capitalizing on the power of neural networks to enhance classification performance. Our comprehensive classifier evaluation encompasses a range of models, including Long Short-Term Memory (LSTM) networks and XGBoost. Remarkably, our results reveal exceptional performance with the XGBoost classifier, achieving an overall accuracy of 84.57 % and a macro-F1 score of 78.21 % on the Sleep-EDF expanded dataset, and an overall accuracy of 86.02 % and a macro-F1 score of 81.96 % on the ISRUC-Sleep dataset. Class-specific accuracies highlight its proficiency, particularly in detecting wake and N2 stages, solidifying its superiority among the classifiers tested. This amalgamation of feature sets, complemented by Principal Component Analysis (PCA) for dimensionality reduction, underscores its significance in yielding top-tier classification outcomes. The integration of traditional feature extraction methods with neural networks as feature extractors creates a robust and comprehensive system for sleep stage classification, offering the advantages of both approaches to enhance the accuracy and reliability of the results.

1. Introduction

Sleep staging is a critical task in sleep medicine, involving the classification of sleep stages based on the analysis of various physiological signals such as electroencephalography (EEG), electromyography (EMG), and electrooculography (EOG) [1]. These signals are typically recorded using multiple sensors placed on the scalp, face, and limbs, collectively forming a polysomnogram (PSG) used for diagnosing and treating sleep disorders [2]. Rechtschaffen and Kales (R&K) [3] and the American Academy of Sleep Medicine (AASM) [4] provide guidelines for sleep staging. Differences between these criteria [3–5] include the mapping of R&K stages (S1, S2, S3, S4) to AASM stages (N1, N2, and N3). In AASM, deep sleep (N3) combines S3 and S4 stages from R&K. Notably, AASM does not include the movement time stage. The EEG serves as a fundamental signal for monitoring brain activity and diagnosing sleep disorders [6]. The electrical activity detected by electrodes positioned on the brain's surface primarily represents the combined

effects of excitatory and inhibitory postsynaptic potentials in apical dendrites of pyramidal neurons located in the superficial cortical layers. Significant synchronous activation of extensive cortical areas is required to generate detectable changes in the electrical potential registered by scalp electrodes [6,7]. Each stage of sleep corresponds to distinct patterns of brainwave activity and neuronal behavior [6]. There are five sub-frequency bands of brainwaves: Delta (1–4 Hz), Theta (4–8 Hz), Alpha (8–13 Hz), Beta (13–30 Hz), and Gamma (30–100 Hz) [8]. EEG recordings during sleep can be assessed by analyzing these frequency bands, along with the observation of sleep-related behaviors like K complexes and spindle waves, enabling the determination of sleep stages [6,9]. EOG is a valuable tool in sleep stage classification, especially during rapid eye movement (REM) sleep. EOG signals aid in distinguishing between REM and non-REM (NREM) sleep, and their analysis, combined with observations of behaviors like K complexes and spindle waves, contributes to the delineation of various sleep stages [10,11].

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Manual analysis, which requires an expert to visually classify each segment, can be a labor-intensive task and is susceptible to human errors [12–14]. These challenges have spurred researchers to devise methods and algorithms for automating sleep-related analyses. These automated approaches offer the benefits of increased speed and enhanced accuracy in sleep stage classification [15].

Numerous techniques for automating sleep studies have been proposed, evolving significantly over the years. Traditional methods typically involve extracting specific signal features from biosignals such as EEG, EOG, and EMG, and subsequently inputting these features into classifiers. For instance, Pascualvaca et al. [16] presented an enhanced sleep stage classification model utilizing advanced intelligent methods, incorporating EOG and EMG signals, and evaluating different SVM multi-classification techniques, which demonstrated significant improvements in classification accuracy. Ghimatgar et al. [17] introduced a single-channel EEG-based scoring method utilizing Modified Graph Clustering Ant Colony Optimization (MGCACO) for feature selection, Random Forest (RF) for classification, and Hidden Markov Model (HMM) for post-processing, achieving high accuracy and robustness on multiple datasets. Shen et al. [18] proposed an algorithm employing Improved Model-Based Essence Features (IMBEFs) and Bagged Trees for classification, demonstrating high accuracy on the Sleep-EDF and DREAMS datasets. More recently, Pei et al. [19] developed a method using Mel-frequency cepstral coefficients (MFCC) features and a deep CNN-LSTM model, demonstrating effectiveness and computational efficiency on the SHHS and UCDDDB datasets.

Deep learning-based models have revolutionized the field by automatically extracting features from raw data, showing significant improvements in classification accuracy. Supratak et al. [20] introduced DeepSleepNet, a model combining CNNs for feature extraction and Bi-LSTMs for capturing temporal dependencies, setting a benchmark on the MASS and Sleep-EDF datasets. Niroshana et al. [21] developed a CNN-GRU model using EEG and EOG signals, achieving high accuracy for both patients and healthy subjects. Jiang et al. [22] proposed a robust classification method using multimodal signal decomposition and an HMM-based refinement process, demonstrating superior performance on the Sleep-EDF and MASS databases. Huang et al. [23] introduced a single-channel EEG sleep staging method using a transition optimized Hidden Markov Model (TO-HMM), incorporating feature extraction with power spectral density (PSD) feature selection and a Gaussian mixture model-Hidden Markov Model (GMM-HMM). Eldele et al. [24] proposed AttnSleep, featuring a multi-resolution CNN and an adaptive feature recalibration (AFR) module, significantly enhancing feature extraction and classification accuracy. Jia et al. [25] introduced GraphSleepNet, an adaptive spatial-temporal graph convolutional network that dynamically learns intrinsic EEG channel connections, achieving superior performance on the MASS dataset. Olesen et al. [26] developed a deep residual network for sleep stage classification using 15,684 PSG studies from five large-scale cohorts, employing a modified ResNet-50 architecture combined with a bidirectional gated recurrent unit (GRU). Guillot and Thorey [15] presented RobustSleepNet, a transfer learning model leveraging an attention mechanism and multi-head recombination, achieving high robustness and generalization across diverse datasets. You et al. [27] developed a method combining time-frequency and fractional Fourier transform (FRFT) domain features with a bidirectional LSTM network, demonstrating high accuracy with fewer parameters. Other models have continued to push the boundaries of sleep stage classification. BootstrapNet [28] employs contrastive learning on raw single-channel EEG data, enhancing feature representation and improving generalization across datasets, though it faces increased computational demands and a dependency on large annotated datasets. CCRRSleepNet [29] integrates convolutional neural networks with relational inductive biases, capturing complex data relationships and achieving high accuracy for single-channel EEG data; however, the added complexity and computational requirements pose challenges for scalability. SingleChannelNet [30] focuses on efficiency with fewer

parameters, making it suitable for resource-constrained devices, though its simplicity may limit generalization without further fine-tuning. SleepPyCo [31] leverages a feature pyramid network and contrastive learning to capture multi-scale features, enhancing accuracy but increasing computational requirements and training complexity. TinySleepNet [32] balances efficiency and performance with fewer parameters and data augmentation techniques, making it suitable for embedded devices, though it might limit the depth of feature extraction. Advanced models such as 3DSleepNet [33] integrate multiple convolutional layers and an attention mechanism to capture spatial-temporal and spatial-spectral features from multi-channel biosignals, combining 3D-CNN and LSTM layers for enhanced sleep stage classification for both healthy and sleep-disorder subjects, demonstrating high accuracy and robust performance. Similarly, MixSleepNet [34] leverages 3D-CNN and Graph Convolutional Networks (GCN) to capture complex spatial-temporal features, dynamically updating the adjacency matrix to reflect changing brain region connections, achieving superior performance across datasets. EEGSNet [35] converts raw EEG signals into spectrograms using Fourier transform, applying CNNs for local feature extraction and Bi-LSTMs for capturing temporal dependencies, excelling particularly in handling unbalanced data and demonstrating strong generalization across different subjects and datasets. The Multilevel Temporal Context Network (MLTCN) [36] integrates Temporal Convolutional Networks (TCN) and Hidden Markov Models (HMM) to capture multilevel temporal features, significantly improving classification accuracy. The Neural Architecture Search (NAS) [37] framework automates the design of optimal neural network architectures for sleep stage classification, utilizing differentiable architecture search (DARTS) and progressive differentiable architecture search (P-DARTS) to optimize feature extraction and classification performance, showcasing high accuracy and generalization. The LSTM-Ladder Networks (LLN) model [38] combines Ladder Network's reconstruction capabilities with LSTM's sequential data handling to enhance classification performance, especially by leveraging both labeled and unlabeled data. Multi-View Spatial-Temporal Graph Convolutional Networks (MSTGCN) [39] use functional and spatial distance-based brain graphs to perform spatial-temporal graph convolutions, capturing both spatial and temporal features effectively, including adversarial domain generalization to enhance robustness across different subjects. CTCNet [40] integrates CNN, Transformer, and Capsule Network technologies, excelling in handling class-imbalanced datasets and significantly improving classification accuracy, particularly for the N1 sleep stage. Jumping Knowledge-Based Spatial-Temporal Graph Convolutional Networks (JK-STGCN) [41] combine adaptive graph learning and jumping knowledge networks to capture spatial and temporal dependencies, resulting in high classification accuracy and efficiency suitable for real-time applications. Finally, the 1D-ResNet-SE-LSTM model [42] integrates one-dimensional residual convolutional neural networks with squeeze-and-excitation modules and LSTM networks, effectively recalibrating feature responses and capturing both spatial and temporal dependencies, demonstrating robustness and high performance across different datasets. In the broader context of time series classification, some models designed for general applications beyond sleep staging also offer valuable insights. Xiao et al. [43] introduced the Robust Temporal Feature Network (RTFN) for time series classification, combining a Temporal Feature Network (TFN) with an LSTM-based attention network (LSTMaN) to capture both local and global temporal features. Ji et al. [44] proposed a space-embedding strategy for anomaly detection in multivariate time series (SES-AD), using LSTM models to process dissimilarity vectors derived from projected time series data, achieving robust anomaly detection. Although these general time series models are not specifically designed for sleep stage classification, they offer methodologies that are applicable to various time-dependent data analysis tasks.

1.1. Advantages and disadvantages of state-of-the-art models

Recent advancements in sleep stage classification have led to the development of several state-of-the-art (STOA) models that leverage deep learning techniques. These models have significantly improved the accuracy and robustness of sleep stage classification by integrating various biosignals, such as EEG, EMG, and EOG. For instance, models like 3DSleepNet [33] and MixSleepNet [34] have demonstrated the efficacy of multi-channel integration and the use of attention mechanisms to enhance focus on critical temporal and frequency information. The inclusion of pseudo-3D convolutions in these models effectively reduces computational complexity while maintaining their ability to extract complex features. However, the computational demands and complex architectures of these models pose significant challenges for real-time applications and implementation. These models often require powerful hardware and optimized software environments, making them less accessible for widespread clinical or at-home use. Other models, such as EEGSNet [35] and the Multilevel Temporal Context Network (MLTCN) [36], focus on spectrogram-based feature extraction and temporal convolutional networks to capture intra-epoch and inter-epoch temporal features. These approaches, while effective in improving classification performance, substantially increase the computational load due to the high dimensionality of spectrogram images and the multiple stages of temporal learning required. Furthermore, these models may face difficulties generalizing across different datasets, necessitating large amounts of data for training and validation to avoid overfitting and to ensure robust performance. Incorporating Long Short-Term Memory (LSTM) networks, models such as the LSTM-Ladder Network [38] and various hybrid models, including 1D-ResNet-SE [42] and LSTM combinations, excel at capturing temporal dependencies and transition rules among sleep stages. However, these models often involve sophisticated layers and loss functions to handle class imbalances, which complicates the training process. Additionally, the reliance on high-quality labeled data can limit their applicability, as acquiring such data is often time-consuming and expensive. Graph convolutional networks (GCNs) have also been adapted for sleep stage classification, as seen in models like the Multi-View Spatial-Temporal Graph Convolutional Networks [39] and Jumping Knowledge-Based Spatial-Temporal Graph Convolutional Networks (JK-STGCN) [41]. These models effectively capture spatial and temporal dependencies by constructing functional and spatial brain graphs and employing advanced graph learning techniques. Despite these benefits, the high computational demand and complexity of implementation associated with these methods present substantial barriers. These models often require extensive preprocessing and optimization, which can be resource-intensive and may limit their practicality in real-time or resource-constrained environments. Lastly, the CTCNet [40] model, which integrates CNN, Transformer, and Capsule Network technologies, showcases the potential for capturing local features, global temporal context, and spatial relationships within EEG data. This model achieves high classification accuracy, particularly in handling class-imbalanced datasets, but its complexity and resource requirements remain significant challenges. The integration of multiple advanced techniques in a single model leads to increased computational overhead, making it difficult to deploy such models in environments with limited computational power.

Overall, while these state-of-the-art models demonstrate remarkable advancements in sleep stage classification by leveraging various deep learning techniques, their high computational demands, complex architectures, and reliance on extensive datasets highlight the need for further research. Optimization for practical, real-time applications remains a critical challenge. Addressing these issues will be crucial for the broader adoption of these models in clinical and home settings, ensuring that the benefits of these advancements can be effectively translated into practical tools for sleep analysis.

1.2. Challenges of existing methods

Despite significant advancements, several challenges continue to impede the development of highly accurate, robust, and generalizable sleep stage classification models. Variability in data quality is a major issue, as EEG signals are prone to noise and artifacts due to different acquisition methods. This variability complicates the creation of universal models that perform consistently across various datasets and settings. Additionally, class imbalance within sleep stage datasets, where certain stages are more prevalent than others, biases training processes and negatively impacts the performance of models on minority classes. This imbalance makes it difficult for models to accurately classify less common stages, leading to potential inaccuracies in sleep stage predictions. Another significant challenge is domain shifts, where differences in data distributions across datasets cause models trained on one dataset to perform poorly on others. Techniques such as ADAST [45] and RobustSleepNet [15] have been developed to address this issue, but domain adaptation remains a significant hurdle. Accurately capturing temporal dependencies in sleep data is also crucial yet challenging. Models like DeepSleepNet [20] and CNN-GRU [21] leverage recurrent neural networks to address this, but fully capturing the complex temporal patterns in sleep stages remains difficult. Effective feature extraction is critical for accurate classification, but both traditional and deep learning methods face limitations. These methods often require extensive labeled data and significant computational resources, which can be prohibitive. The high computational demands of deep learning models further limit their use in resource-constrained environments, such as home-based monitoring systems, where lightweight and efficient models are needed. Inconsistent evaluation metrics and protocols across studies also hinder direct performance comparisons, underscoring the need for standardized benchmarks in the field. Ensuring model interpretability and clinical relevance is essential, as many models function as black boxes, making it difficult for clinicians to trust and understand their decisions. This lack of transparency can be a barrier to clinical adoption. Additionally, handling class imbalance and domain adaptation effectively remains a significant challenge. Models like CoSleepNet [46] address class imbalance in EEG-EOG datasets but face difficulties in generalizing across varied datasets. Similarly, personalized approaches like LightSleepNet [47] aim to mitigate domain adaptation issues, but this area requires further refinement. Other challenges include ensuring scalability and robustness in real-time applications. Handling large-scale sleep datasets efficiently and processing data in real-time for applications like home monitoring and wearable devices are crucial. Integrating data from various biosignals and other sources, such as heart rate and respiration, can enhance classification accuracy but also increases model complexity and computational requirements. Advanced techniques such as transfer learning and domain adaptation offer promising solutions but still face challenges related to overfitting and generalization. Models must be robust to noise, particularly in non-clinical environments where various noise sources can degrade performance. Interoperability between different systems and standardization of data formats and protocols are also critical for broader application and comparison of models. Lastly, ensuring explainability and trust in model decisions is vital, not just for model interpretability but also to provide accurate and meaningful explanations in a clinical context.

In this paper, we propose a unique approach to sleep staging that emphasizes the comprehensive amalgamation of various features and signal modalities. Our method leverages different modalities, including two distinct EEG channels and one EOG channel, alongside a fusion of diverse techniques such as Short-Time Fourier Transform (STFT), discrete wavelet transform, and CNN for feature extraction. By amalgamating all these features and streamlining the feature vector dimensionality using Principal Component Analysis (PCA), we arrive at a holistic solution for sleep stage classification. Our source code is publicly available at https://github.com/mahdisamaee/Multi-modal-Signal-Integration-for-Enhanced-Sleep-Stage-Classification_new.git.

The rest of the paper is organized as follows: Section 2 provides an in-depth exploration of the materials and methods, shedding light on our dataset and the techniques we employed. Moving forward, Section 3 delves into the comparison metrics we've utilized, while presenting the results of our experiments. Section 4 opens the floor for a detailed discussion of our findings. Finally, in Section 5, we summarize our work with a conclusion that encapsulates the key takeaways and implications of our research.

2. Materials and methods

2.1. Used database

In this study, we employed two key datasets: the Sleep-EDF Database Expanded (version 1.0.0)¹ [48,49] and the ISRUC-Sleep dataset² [50]. The Sleep-EDF database comprises 197 comprehensive whole-night PSG sleep recordings, including crucial physiological signals such as EEG, EOG, chin EMG, and event markers. We exclusively utilized the "Sleep Cassette" subset. The ISRUC-Sleep dataset includes sleep recordings from 118 subjects, recorded over multiple nights, with each recording containing comprehensive PSG data, including EEG, EOG, EMG, ECG, and airflow signals. For our study, we specifically utilized one EOG channel and two EEG channels from both datasets. This approach ensures a balance between data richness and computational efficiency.

2.2. Data preparation

In this study, data preparation involved segmenting the signals into discrete 30-s epochs, as the expert annotations in the dataset are provided in these intervals. The Sleep-EDF dataset is sampled at 100 Hz, while the ISRUC-SLEEP dataset is sampled at 200 Hz. To ensure consistency, we downsampled the ISRUC-SLEEP dataset to 100 Hz. The distribution of the 30-s sleep signal samples following this preparation is detailed in Tables 1 and 2.

2.3. Methodology

In this section, we present a rigorous methodology for sleep stage classification. Our approach is systematically structured into five key phases, each with distinct contributions to the overall classification process: Elementary Feature Extraction, Secondary Feature Extraction, Feature Concatenation, Dimension Reduction, and Classification. The visual representation of our methodology is depicted in Fig. 1.

Table 1

Distribution of 30-s sleep signal sets (Sleep-EDF Expanded dataset).

Sleep stages	awake	N1	N2	N3	REM	Total
Number of each 30 s sample	65,795	21,469	68,633	12,991	25,767	194,655

¹ This dataset is available on the PhysioNet website at <https://physionet.org/content/sleep-edfx/1.0.0/>. The 'Sleep Cassette' portion of this dataset, used in this study, contains 153 PSG recordings from subjects aged 25–101. Two PSGs, each approximately 20 h long, were recorded during two consecutive day-night periods at the subjects' homes. The EOG and EEG signals were sampled at a rate of 100 Hz.

² The ISRUC Sleep Dataset is available on <https://sleeptight.isr.uc.pt>. This dataset includes PSG recordings from 118 subjects aged 20–85. The EEG signals were sampled at a rate of 200 Hz. The recordings were conducted in a controlled environment at the Sleep Medicine Centre of the Hospital of Coimbra University (CHUC), providing high-resolution data for detailed sleep stage analysis.

Table 2

Distribution of 30-s sleep signal sets (ISRUC-SLEEP dataset).

Sleep stages	awake	N1	N2	N3	REM	Total
Number of each 30 s sample	23,945	14,401	35,034	21,871	14,252	109,503

2.3.1. Signal modalities

Our investigation into sleep stage classification incorporates three essential signal channels: horizontal EOG, and two EEG channels. These channels collectively provide unique physiological insights essential for gaining a comprehensive understanding of the intricacies of sleep dynamics.

2.3.2. Feature extraction approaches

In the Elementary Feature Extraction phase, we employ a trinity of fundamental methodologies to extract informative features. These methodologies include the STFT and the Discrete Wavelet Transform (DWT). This multifaceted approach is characterized by a purposeful synergy that enhances the effectiveness of our classification methodology. The STFT reveals the temporal evolution of frequency components, enriching our understanding of how spectral content varies over time. This, in turn, enables the discernment of dynamic changes and trends within sleep stages. On the other hand, the DWT, by dissecting signals into time-scale components, uncovers fine-grained details and abrupt shifts that may carry significant discriminants for sleep stages. Both the STFT and the DWT, with their distinct perspectives, work in tandem to provide a comprehensive frequency domain analysis of the signal.

2.3.2.1. STFT calculation for each signal modality. We selected four distinct window sizes for Short-Time Fourier Transform (STFT) calculations: 200, 400, 500, and 800 samples, corresponding to 2, 4, 5, and 8 s of signal data at a 100 Hz sampling frequency. This selection balances temporal and frequency resolution, ensuring comprehensive signal analysis. Shorter windows of 200 and 400 samples provide higher temporal resolution, crucial for capturing rapid signal changes. These include transient events like sleep spindles (11–16 Hz), which typically last about 0.5–2 s, and brief oscillatory patterns characteristic of lighter sleep stages (N1 and N2). This allows for precise detection and analysis of these short-duration phenomena. Longer windows of 500 and 800 samples offer superior frequency resolution, essential for accurately detecting slower oscillations and more stable patterns, such as delta waves (<4 Hz), which dominate in deep sleep (N3). These longer windows also help in capturing the mixed frequency bands characteristic of REM sleep, including both low-frequency theta waves and higher-frequency activities. Our selection of these four window sizes was validated through testing different window sizes and determining that these provided the most reliable and accurate classification results across all sleep stages. This approach ensures that we effectively capture the diverse frequency patterns and temporal dynamics inherent in sleep signals, enhancing the robustness and accuracy of our sleep stage classification model. We have also scrutinized overlap sizes of 60 %, 70 %, and 80 %, with a compelling outcome - a 70 % overlap emerged as the optimal choice which is in accordance with prior research [51]. The window type is another parameter to consider when calculating the STFT. Our extensive experimentation has shown that, although we explored various window types, they led to negligible alterations in our results. As a result, we have chosen the Hamming window as our preferred option for analysis. A visual depiction of this method is illustrated in Fig. 2.

2.3.2.2. Discrete wavelet transform calculation for each signal modality. In parallel, the DWT plays a central role in dissecting signals into time-scale components. We systematically apply the DWT across six distinct levels and concatenate the outputs, thereby unveiling intricate high- and low-frequency features at various scales. This multi-tiered approach enriches

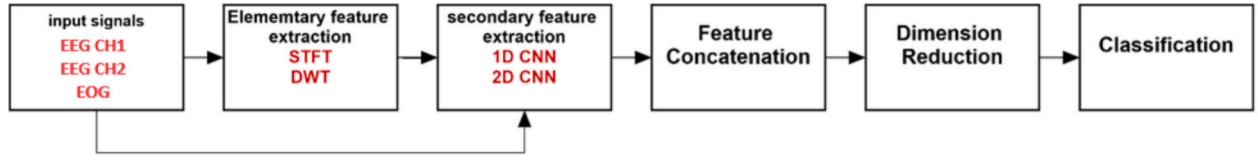


Fig. 1. Proposed method steps.

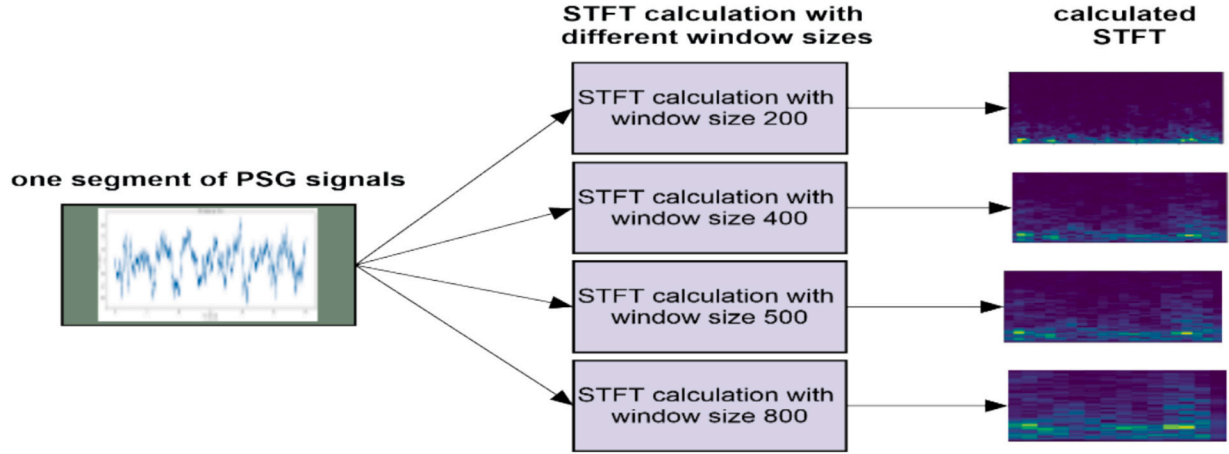


Fig. 2. Calculating STFT with 4 window sizes for each signal modality.

the precision and dependability of our classification process, empowering us to capture and decode the complexities of the signals. A visual depiction of this method is illustrated in Fig. 3.

2.3.3. Applying convolutional neural networks

Our Secondary Feature Extraction phase uses CNN to extract features from diverse data sources, namely raw signal data, STFT results, and wavelet-transformed data.

2.3.3.1. Utilizing one-dimensional convolutional neural networks as feature extractors. In our relentless pursuit of feature extraction excellence, we've harnessed the power of a tailored one-dimensional CNN architecture. This network's foundational structure draws inspiration from [20,52], and we have meticulously adapted and fine-tuned it to align with our specific requirements. This architecture plays a pivotal role in capturing salient features from both RAW signal data and wavelet-transformed data, ensuring a comprehensive extraction of essential information from these two distinct signal sources. The visual representation of this architecture is depicted in Fig. 4. It is crucial to emphasize that our ultimate goal is to harness these networks as feature extractors, rather than as the final classifiers. To shed light on the intricacies of this architecture, let's delve into the detailed training process:

2.4. Multi-kernel 1d cnn for dual data streams

We've previously established our intention to employ this 1D CNN structure for both RAW signal data and wavelet-transformed data. A significant aspect of this architecture is the consideration of four different kernel sizes for the input convolutional layer. This corresponds to having four distinct networks for each signal modality and data type (wavelet features and RAW signal), with each network utilizing varying input kernel sizes.

2.5. Diverse data handling

We allocate distinct networks for each set of data. For every signal modality, including horizontal EOG and two EEG channels, individual networks are designated. This segregated approach is applied to both the raw data and wavelet features within each signal modality.

2.6. Two-stage training

An important facet of this architecture is our two-stage training process. Initially, we train this structure with our train data, focusing on the five-class output. Upon the completion of this training phase, we

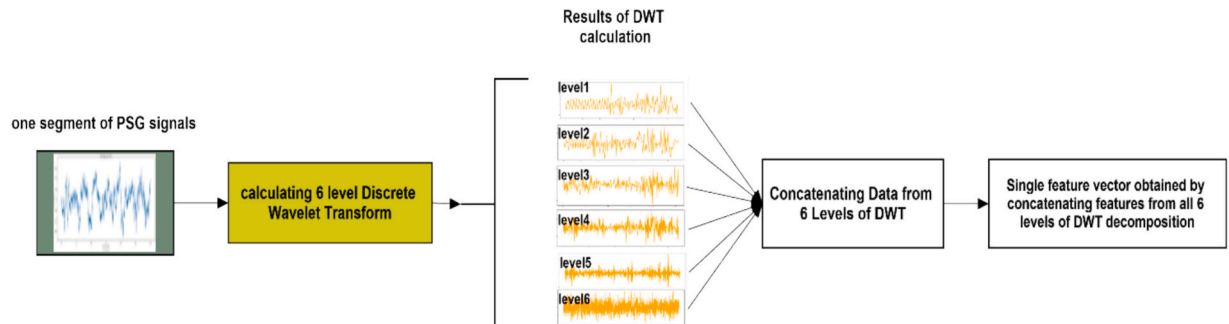


Fig. 3. Calculating DWT in 6 levels for each signal modality.

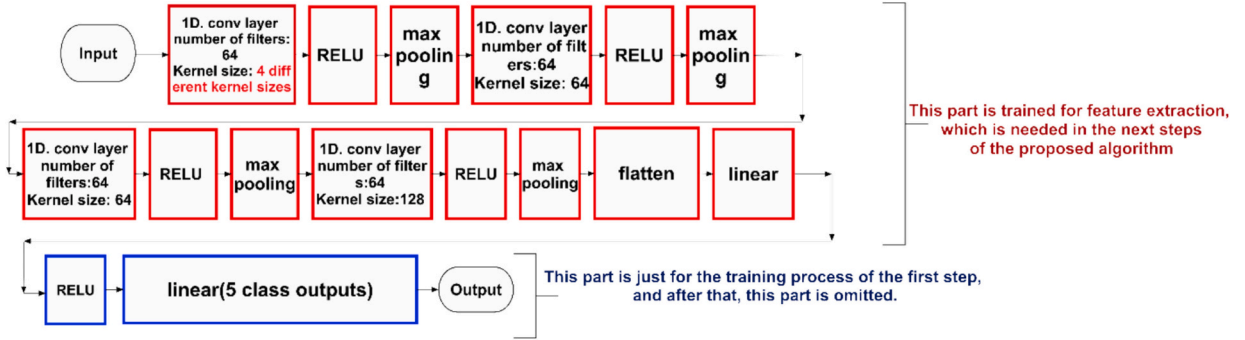


Fig. 4. Proposed 1D CNN.

strategically omit the last linear layer, transforming the remaining components into feature extractors rather than serving as the final classifier.

By adopting this methodological approach, we ensure that our networks are finely tuned to function as feature extractors, amplifying the richness and depth of information we extract from our diverse signal sources. Kernel sizes that we consider for the first convolutional layer are 50, 100, 200 and 400. In the final layer of the initial network segment, which is a linear layer, we have configured it with 128 nodes. The number of filters and kernel sizes for each convolutional layer is depicted in Fig. 5. The optimizer has been set to Adam with default parameters. Specifically, the learning rate, beta1, and beta2 values are configured as 0.001, 0.9, and 0.999, respectively, which are PyTorch's default values and have proven effective for our specific application.

2.6.1. Employing two-dimensional convolutional neural network as feature extractors

In a simultaneous endeavor, we introduce a two-dimensional CNN architecture used to capture the intricate temporal-frequency dynamics inherent in STFT data. The visual representation of this architecture is depicted in Fig. 5. The figure displays the number of filters and kernel sizes for each convolutional layer, while the optimizer and its parameters are consistent with those previously mentioned for the 1D CNN. Additionally, for the linear layer, we have retained 128 nodes as in the previous configuration.

As previously mentioned, in the context of one-dimensional CNNs, we uphold a consistent approach for two-dimensional CNNs. For every signal modality and for each set of STFT features, a dedicated 2D CNN is employed. Initially, these networks are trained with a primary focus on the five-class output. Following this initial training phase, a strategic modification is implemented. The last linear layer is deliberately omitted, converting the remaining components into feature extractors, thereby repositioning these networks from final classifiers to feature extraction engines. This two-stage approach ensures that the networks

are optimally configured to unravel the intricate temporal-frequency dynamics of the STFT data while enhancing the depth and richness of feature extraction. A visual depiction of this method is illustrated in Fig. 6.

2.6.2. Feature concatenation and dimension reduction

The Feature Concatenation phase involves the merging of CNN outputs from STFT, wavelet transform, and raw data sources. This amalgamation results in a comprehensive feature vector with a high-dimensional representation of our data. PCA is subsequently applied to reduce dimensionality to enhance efficiency and generalization. This phase strategically preserves the most informative aspects of the data while reducing the high-dimensional feature space. A visual depiction of this method is illustrated in Fig. 7.

Fig. 8 shows a complete diagram of the proposed method.

2.6.3. Classification

In the final classification phase, we rigorously examine the performance of various classifiers, including Random Forest, Decision Tree, Adaboost, Linear Discriminant Analysis, Logistic Regression, Quadratic Discriminant Analysis, Multi-Layer Perceptron (MLP), and Long Short-Term Memory (LSTM). This comprehensive exploration provides valuable insights into the complex realm of sleep stage classification, highlighting the intricate interplay between feature extraction, dimensionality reduction, and classification techniques. We evaluated several traditional machine learning classifiers, such as Random Forest, Decision Tree, Adaboost, Linear Discriminant Analysis, Logistic Regression, Quadratic Discriminant Analysis, and XGBoost. These classifiers were applied to the extracted features to assess their performance in sleep stage classification. For these traditional classifiers, we employed a random grid search approach for hyperparameter training selection. This method involves specifying a range of possible values for each hyperparameter and randomly sampling combinations to identify the best-performing set, ensuring thorough exploration and optimal

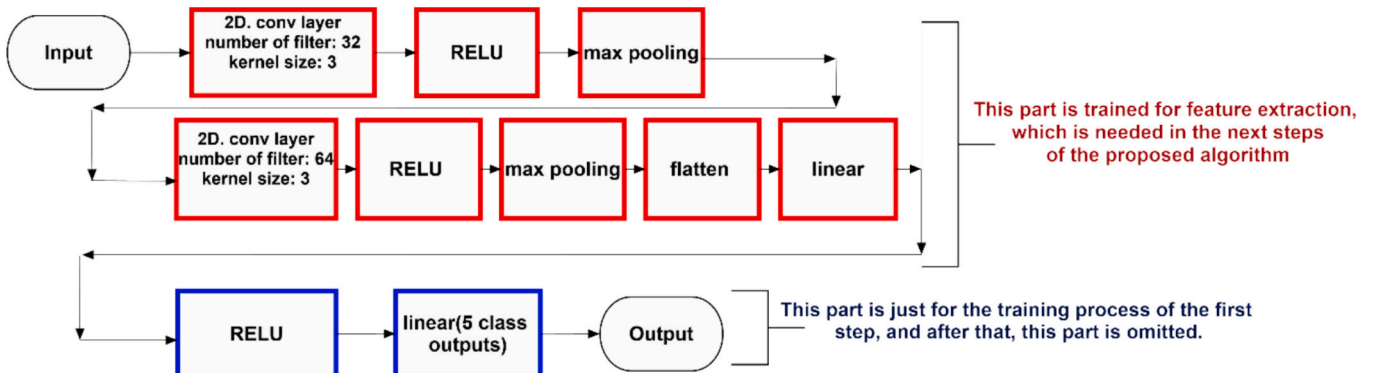


Fig. 5. Proposed 2D CNN.

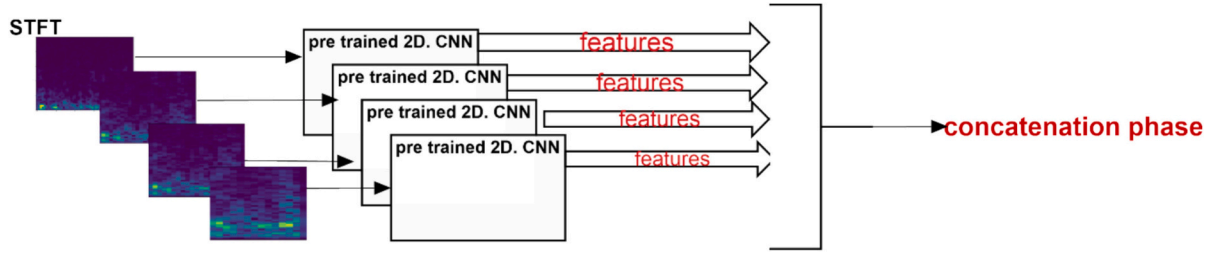


Fig. 6. Implementation of 2D CNN on STFT data.

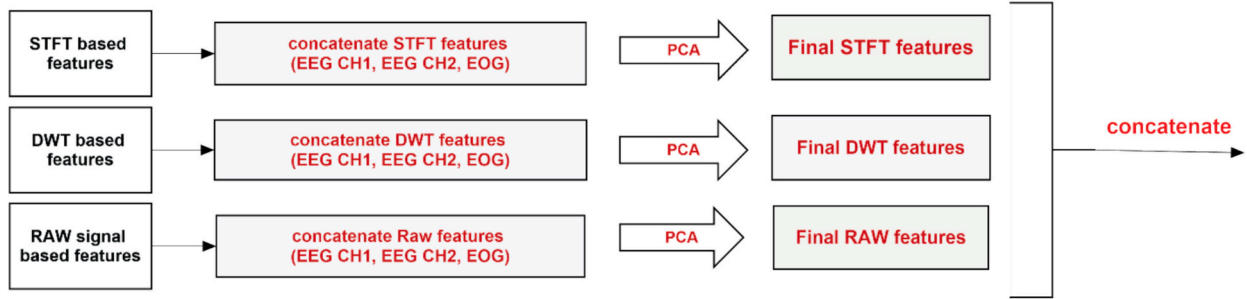


Fig. 7. Concatenating the results of each feature extraction method individually, performing PCA on each set of features, and then concatenating the reduced feature sets.

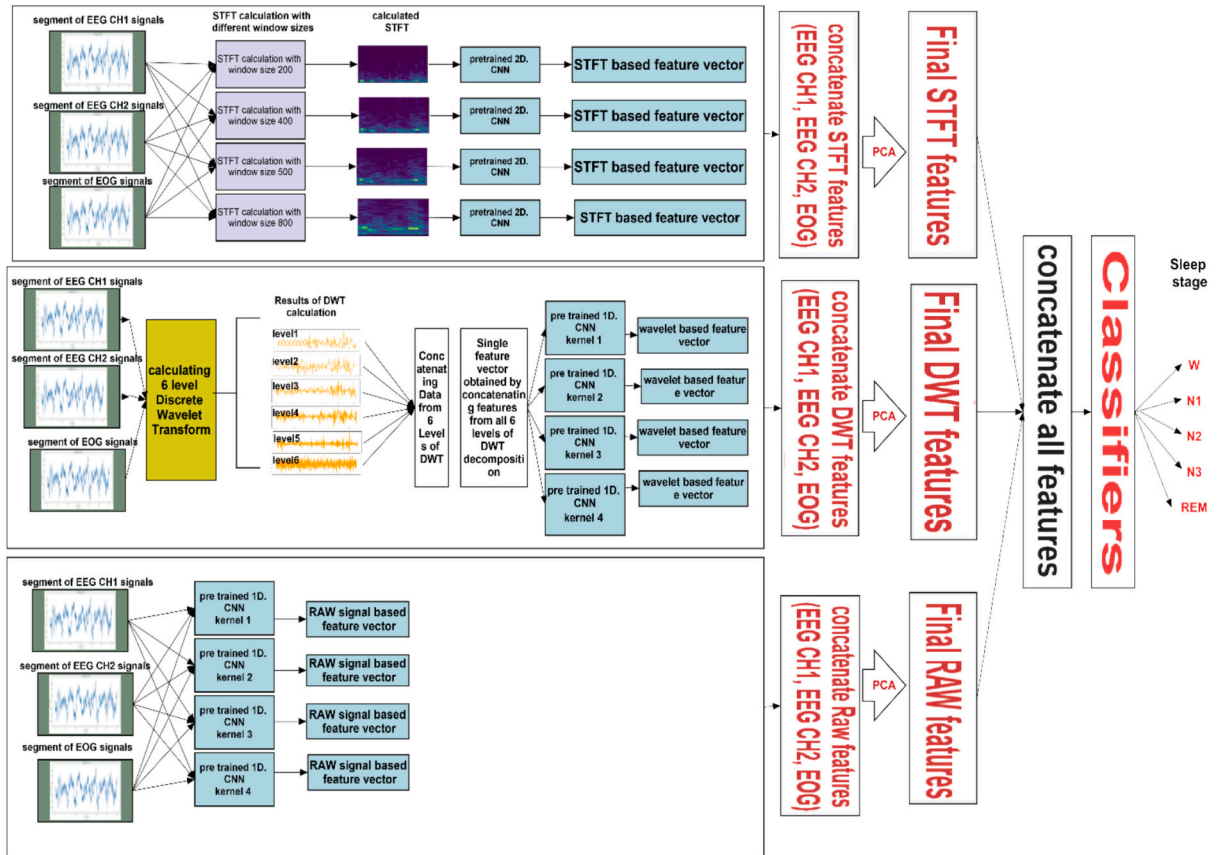


Fig. 8. Diagram of the proposed method.

performance. The MLP classifier was included due to its ability to model complex relationships within the data. The MLP architecture consists of two fully connected layers, supplemented by dropout and batch normalization techniques to prevent overfitting and enhance

generalization. This structure allows the MLP to effectively capture and learn intricate patterns present in the sleep data. To further enhance performance, we designed an LSTM network, particularly well-suited for time-series data due to its capability to capture long-term dependencies.

Our LSTM model consists of a three-layer architecture with 128 hidden units per layer, combined with batch normalization and dropout layers to improve generalization and mitigate overfitting. Each LSTM layer captures long-term dependencies within the sequential EEG data, while the batch normalization layer normalizes the output, and the dropout layer reduces the risk of overfitting. The extracted features from the LSTM are then fed into an MLP classifier, which includes two fully connected layers, each followed by dropout and batch normalization, with the final output layer consisting of five units corresponding to the five sleep stages. For the LSTM network, we tested various configurations, including different numbers of layers, hidden units, dropout rates, and batch normalization setups, ultimately selecting the best-performing architecture. To create the input sequences for the LSTM, overlapping sequences of EEG segments were generated. Each sequence is composed of eight consecutive 30-s EEG segments, ensuring that temporal dynamics are preserved. The sequences are constructed by sliding a window of length eight over the EEG data, where each new sequence starts one segment after the previous one. This method ensures that each segment contributes to multiple sequences, providing a comprehensive view of the temporal context around each sleep stage transition. Various sequence lengths, ranging from 2 to 10, were tested, with the best performance achieved using sequences of length 8.

In summary, our classification phase involves a rigorous evaluation of various classifiers, including traditional machine learning models and advanced neural network architectures like MLP and LSTM, to determine the most effective approach for sleep stage classification. The integration of these diverse classifiers provides a comprehensive understanding of their respective strengths and weaknesses, ultimately leading to the identification of the optimal model for our task.

3. Experimental results

In this section, we present the results of sleep stage classification using various classifiers. We employed a 20-fold cross-validation approach with a subject-wise split to ensure that the same subject does not appear in both the training and validation sets, enhancing the generalization capability of the models across different subjects. Our experiments utilized two datasets: the sleep-edf-database-expanded-1.0.0 and the ISRUC-Sleep database.

The performance metrics, including overall accuracy, MF1, and class-specific accuracies for each sleep stage, are summarized in Tables 3 and 4. With the True Positives (TP_i), False Positives (FP_i), True Negatives (TN_i), and False Negatives (FN_i) for each class i , we can define the overall accuracy (ACC) and macro-F1 (MF1) as follows.

$$ACC = \frac{\sum_{i=1}^K TP_i}{M}, MF1 = \frac{1}{K} \sum_{i=1}^K \frac{2 * Precision_i * Recall_i}{Precision_i + Recall_i}, Precision_i = \frac{TP_i}{TP_i + FP_i}, Recall_i = \frac{TP_i}{TP_i + FN_i}$$

Here, M represents the overall sample count, while K denotes the quantity of classes or sleep stages.

These results demonstrate the performance of different classifiers in classifying sleep stages. Overall accuracy varies across the classifiers,

with the highest overall accuracy achieved by the XGBoost classifier at 84.57 % on Sleep EDF expanded dataset and 86.02 % on ISRUC-SLEEP Dataset. Class-specific accuracies vary significantly between sleep stages. Notably, wake, N2 and N3 sleep stages tend to have higher accuracies across most classifiers, while N1 stage exhibits lower accuracies in general.

We have selected the XGBoost classifier as our top performer. Table 5 presents the standard deviation of measured metrics across different folds. Tables 6 and 7 compare our method's results with those from state-of-the-art papers.

4. Discussion

In this study, we undertook an ambitious effort to harness the complete potential of available signals for sleep stage classification. Our findings illustrate the promising results. We firmly attribute this remarkable performance to our innovative approach of harnessing diverse signal modalities and deploying a range of feature extraction methods to enhance the accuracy of sleep stage classification. This section delves into the scientific rationale behind our approach and sheds light on the strategic combinations that underpin our impressive results. Given the excellent outcomes achieved with the XGBoost classifier, our analysis in this section centers on this particular classifier.

4.1. Discussion on the performance of different classifiers

The performance of various classifiers on the Sleep EDF Extended and ISRUC-SLEEP databases was evaluated to determine their effectiveness in sleep stage classification. The analysis revealed that ensemble methods, particularly XGBoost and Random Forest, consistently achieved high overall accuracy and macro-averaged F1 scores (MF1) across both datasets. This suggests that these classifiers are robust and generalizable, making them suitable candidates for practical applications in sleep stage classification. XGBoost demonstrated the highest overall accuracy on both datasets, with 84.57 % on the Sleep EDF Extended database and 86.02 % on the ISRUC-SLEEP database. Random Forest followed closely, achieving 83.15 % and 85.96 % on the respective datasets. The high performance of these ensemble methods can be attributed to their ability to handle complex patterns and interactions within the data. Their robustness across different datasets indicates their potential for deployment in clinical settings or personal health monitoring systems. Analyzing the class-wise performance, it is evident that the classifiers generally excel in identifying the Wake stage, with LSTM and Random Forest achieving accuracies above 97 % on the Sleep EDF Extended database and ISRUC-SLEEP database. However, the N1 stage remains a challenge across all classifiers, with significantly lower accuracies. This indicates that the characteristics of the N1 stage are less distinct and more difficult to capture, necessitating further research and potential feature engineering to enhance detection accuracy. For the N2 and N3 stages, ensemble methods like Random Forest, XGBoost, and Adaboost performed well, highlighting their ability to manage more stable and distinct sleep stages effectively. Specifically, Adaboost showed strong performance in the N3 stage on the Sleep EDF

Table 3

Results of sleep stage classification using different classifiers on Sleep EDF expanded dataset.

Classifier	Overall Accuracy	MF1	Wake	N1	N2	N3	REM
Random Forest	83.15	76.00	97.28	37.07	93.03	86.51	56.28
XGBoost	84.57	78.21	96.93	38.24	89.38	90.28	74.02
Adaboost	79.36	71.15	91.83	38.71	93.47	90.67	41.53
Decision Tree	77.94	67.66	94.19	31.40	85.48	77.78	50.10
Linear Discriminant Analysis	80.66	72.98	93.72	27.09	79.90	90.48	84.98
Logistic Regression	73.41	51.15	91.44	10.26	72.90	7.74	94.29
Quadratic Discriminant Analysis	68.31	58.60	97.06	36.80	56.61	76.98	19.27
Multi-Layer Perceptron	82.63	74.45	96.72	31.05	86.59	91.87	71.69
LSTM	82.71	75.14	98.07	39.95	89.67	75.00	58.41

Table 4

Results of sleep stage classification using different classifiers on ISRUC-SLEEP Dataset.

Classifier	Overall Accuracy	MF1	Wake	N1	N2	N3	REM
Random Forest	85.96	81.54	97.75	54.37	82.52	90.70	78.86
XGBoost	86.02	81.96	97.20	59.61	81.62	90.22	79.62
Adaboost	84.31	79.74	95.88	52.62	79.82	89.98	79.62
Decision Tree	82.91	77.00	92.97	46.60	81.93	89.13	70.86
Linear Discriminant Analysis	85.76	81.42	94.56	68.16	80.01	90.10	82.67
Logistic Regression	85.07	81.33	95.93	61.94	79.37	89.25	80.38
Quadratic Discriminant Analysis	82.11	77.73	94.67	54.17	69.92	95.41	81.14
Multi-Layer Perceptron	83.98	79.18	97.31	48.93	78.98	86.71	82.67
LSTM	85.10	81.53	96.58	64.33	77.35	89.49	81.64

Table 5

Standard deviation of measured metrics across different folds.

Metric	Overall Accuracy	MF1	Wake	N1	N2	N3	REM
Standard deviation over Folds (for Sleep EDF expanded dataset)	0.0207	0.0220	0.0246	0.0343	0.0292	0.0602	0.0455
Standard deviation over Folds (ISRUC-SLEEP Dataset)	0.0219	0.0202	0.0316	0.0408	0.0444	0.0555	0.0541

Table 6

Comparison of Performance Metrics Obtained by Our Approach with State-of-the-Art Algorithms on Sleep EDF expanded dataset.

	Overall Metrics		Per-Class F1				
	ACC	MF1	W	N1	N2	N3	REM
Ref [35]	83.02	77.26	93.00	45.00	85.00	80.00	84.00
Ref [36]	81.00	74.00	85.95	22.04	87.70	90.68	85.91
Ref [38]	84.30	76.88	91.42	37.66	90.58	75.89	83.82
CTCNet [40]	82.50	79.10	92.50	53.80	86.80	87.30	74.80
Ref. [37]	80.0 %	72.7 %	91.1 %	34.50 %	88.31 %	79.66 %	64.04 %
Proposed Method	84.57	78.21	96.93	38.24	89.38	90.28	74.02

Table 7

Comparison of Performance Metrics Obtained by Our Approach with State-of-the-Art Algorithms on ISRUC-SLEEP dataset.

	Overall Metrics		Per-Class F1				
	ACC	MF1	W	N1	N2	N3	REM
MixSleepNet [34]	83.00	82.10	89.90	62.50	81.90	89.90	86.00
3DSleepNet [33]	83.20	81.40	89.60	59.60	83.20	90.90	83.80
MSTGCN [39]	82.10	80.80	89.40	59.60	80.60	89.00	85.60
JK-STGCN [41]	82.00	79.80	89.50	55.00	81.10	88.30	85.00
Ref [42]	81.97	79.94	88.98	55.53	82.50	89.25	83.46
Proposed Method	86.02	81.96	97.20	59.61	81.62	90.22	79.62

Extended database, achieving an accuracy of 90.67 %. Despite being an ensemble method, Adaboost did not achieve results as high as XGBoost or Random Forest. Adaboost's overall accuracy was 79.36 % on the Sleep EDF Extended database and 84.31 % on the ISRUC-SLEEP database. The relatively lower performance of Adaboost can be attributed to its sensitivity to noisy data and outliers, which can significantly impact its boosting process. Unlike Random Forest and XGBoost, which use multiple weak learners to form a strong model through bagging and gradient boosting techniques respectively, Adaboost iteratively adjusts the weights of misclassified instances. This can lead to overfitting, especially in datasets with high variability like sleep stage data. The ISRUC-SLEEP database generally yielded higher performance metrics compared to the Sleep EDF Extended database. This discrepancy could be due to differences in the number of subjects, variability in sleep patterns, or recording conditions between the datasets. Despite this variability, the consistent top performance of XGBoost across both datasets underscores its effectiveness in managing different sleep data characteristics. In practical terms, the high performance of XGBoost and Random Forest indicates that these models are well-suited for sleep stage classification tasks. Their robustness and high accuracy make them viable options for deployment in clinical diagnostics or consumer sleep monitoring devices. However, the persistent challenge in N1 stage

detection highlights the need for further research, possibly exploring alternative modeling techniques or advanced feature extraction methods to improve classifier performance in this area.

In conclusion, ensemble methods such as XGBoost and Random Forest have demonstrated superior performance in sleep stage classification tasks across multiple datasets. While these classifiers excel in detecting most sleep stages, the N1 stage remains an area needing improvement. Future research should focus on enhancing the detection accuracy of challenging stages and addressing dataset-specific characteristics to further improve the applicability of these models in real-world scenarios. Additionally, understanding the limitations of Adaboost in handling noisy and variable data can inform the development of more robust ensemble methods for sleep stage classification.

4.2. Result comparison with state-of-the-art methods

The performance of the proposed method was evaluated against various state-of-the-art (SOTA) algorithms using two different sleep databases: Sleep EDF Extended and ISRUC-SLEEP. The results demonstrate that the proposed method consistently outperforms existing techniques across several key metrics.

4.2.1. Sleep EDF extended database

The proposed method achieved an accuracy (ACC) of 84.57 %, the highest among all compared methods. This indicates a superior overall performance in correctly classifying sleep stages. The macro F1-score (MF1) of 78.21 % further highlights the method's robustness in handling the imbalanced nature of the dataset, ensuring balanced performance across different sleep stages. When examining the per-class accuracy, the proposed method excels particularly in classifying the Wake stage with an accuracy of 96.93 %, significantly outperforming other methods. For the N1 stage, the method shows competitive performance with an accuracy of 38.24 %, which, while not the highest, demonstrates reasonable accuracy given the inherent difficulty of classifying this stage. The accuracies for N2 and N3 stages are 89.38 % and 90.28 %, respectively, both of which are among the highest, indicating excellent capability in distinguishing these stages. However, for the REM stage, the proposed method's accuracy of 74.02 % is slightly lower than some other methods, suggesting a potential area for further improvement.

4.2.2. ISRUC-SLEEP database

On the ISRUC-SLEEP database, the proposed method again shows superior performance with an accuracy of 86.02 %, the highest among all methods compared. The macro F1-score of 81.96 % confirms the method's effectiveness in providing balanced classification across various sleep stages. For the per-class accuracy, the proposed method achieves impressive results with a Wake stage accuracy of 97.20 %, demonstrating its exceptional capability in correctly identifying this stage. The N1 stage accuracy is 59.61 %, which is competitive and shows a notable improvement compared to other methods. The N2 stage classification performance is strong with an accuracy of 81.62 %, while the N3 stage achieves an accuracy of 90.22 %, indicating reliable performance in identifying deep sleep stages. However, the REM stage accuracy of 79.62 % suggests a slight lag behind some methods, indicating a potential area for further refinement.

4.2.3. Overall comparison

The proposed method exhibits remarkable performance across both databases, demonstrating its generalizability and effectiveness in sleep stage classification tasks. The high accuracy and macro F1-scores across both datasets suggest that the method is robust and well-suited for practical applications in sleep analysis. While the method excels in most stages, the slightly lower performance in the REM stage presents an opportunity for future improvements. Enhancements in this area could further solidify the method's standing as a leading approach in sleep stage classification.

4.3. Methodology comparison with state-of-the-art approaches

In this section, we present a comparative analysis of our proposed sleep stage classification methodology against several state-of-the-art approaches, with a particular focus on feature extraction and classification techniques. This comparison highlights the advantages of our methodology in capturing both temporal and spectral features from raw data, as well as its efficacy in leveraging deep learning architectures for robust classification. Additionally, we emphasize the strategic combination of traditional feature extraction methods with neural networks and the use of relatively simple neural network architectures. Our methodology integrates Short-Time Fourier Transform (STFT), Discrete Wavelet Transform (DWT), and features from raw signals for elementary feature extraction from horizontal EOG, and two EEG channels. This trifold approach offers a comprehensive analysis of frequency components, capturing both transient and persistent patterns in the signals. The use of multiple window sizes in STFT allows for detailed temporal resolution, while DWT provides multi-scale decomposition, enabling the detection of subtle signal variations. Additionally, extracting features from raw data ensures that no potentially valuable information is lost

during preprocessing. Comparatively, the methodology employed in 3DSleepNet [33] uses 3D Convolutional Neural Networks (3D-CNN) and Pseudo-3D convolutions to extract spatial-temporal features from multi-channel biosignals. While effective, this approach primarily focuses on spatial-temporal interactions without the granular frequency analysis provided by DWT and STFT in our method. Similarly, MixSleepNet [34] leverages both 3D-CNN and Graph Convolutional Networks (GCN) to capture spatial and spectral relationships. However, our use of STFT, DWT, and raw signal features offers a more refined frequency domain analysis, which is critical for distinguishing subtle sleep stage transitions. The EEGSNet model [35], which transforms EEG signals into spectrograms for CNN-based feature extraction, relies heavily on the visual representation of spectral content. Although effective, it lacks the temporal adaptability achieved through our multi-window STFT approach. Furthermore, our method's combination of 1D and 2D CNNs for secondary feature extraction ensures that both raw signal and frequency-transformed data are thoroughly analyzed, providing a richer feature set than the single-domain focus of EEGSNet. Our approach stands out by combining traditional feature extraction methods with neural network architectures, specifically 1D and 2D CNNs, which are relatively simple compared to the more complex architectures used in some state-of-the-art methods. For example, the CTCNet [40] and other hybrid models incorporate advanced modules like Transformers and Capsule Networks. While these models are powerful, they often require substantial computational resources and extensive training. Incorporating Long Short-Term Memory (LSTM) networks, our methodology captures long-term dependencies and transition rules among sleep stages, similar to the approaches used in other models [40,42]. Moreover, advanced methodologies like the Multi-View Spatial-Temporal Graph Convolutional Networks (STGCN) [39,41] focus on spatial relationships between EEG channels through graph-based learning. While these approaches excel in capturing spatial dependencies, they often overlook the detailed frequency domain analysis provided by our combined STFT, DWT, and raw data approach. Our methodology's robust classification phase evaluates multiple classifiers, including Random Forest, Decision Tree, Adaboost, MLP, and LSTM, ensuring a comprehensive understanding of classifier performance across different feature sets. This contrasts with the single-model focus of many other studies, allowing us to identify the optimal classifier for our specific application.

In summary, our method offers a balanced and thorough analysis by integrating sophisticated feature extraction techniques with advanced classification models. By capturing intricate temporal and spectral features from raw signals, STFT, and DWT, and leveraging relatively simple yet effective neural network architectures, our approach enhances classification accuracy and robustness, setting it apart from existing state-of-the-art methodologies. The strategic combination of traditional and neural network-based feature extraction methods ensures a comprehensive and efficient analysis, making our method both innovative and practical.

4.4. Discussion on computational cost

Our proposed method exhibits significant computational costs primarily due to the extensive feature extraction stage involving a suite of neural networks tailored for three different signal modalities. Specifically, we employ 12 neural networks for each modality, comprising 4 one-dimensional (1D) convolutional neural networks (CNNs) for feature extraction from both raw and wavelet-transformed data, and 4 two-dimensional (2D) CNNs for feature extraction from Short-Time Fourier Transform (STFT) images. This setup results in a total of 36 neural networks across all modalities. The individual parameter counts for the single-layer neural networks are substantial: the 1D CNNs range from 392,369 to 1,271,089 parameters depending on the kernel size, while each 2D CNN encompasses 2,477,189 parameters. Collectively, these networks comprise a total of 24,976,167 parameters. The most computationally demanding component is the 2D CNN, which requires

approximately 25 s per epoch on our hardware configuration (CPU: 12th Gen Intel(R) Core i7-12700K, GPU: GeForce RTX 3060). Given the independent nature of our networks, parallel execution is not mandatory; hence, we perform sequential feature extraction. In this sequential approach, the maximum number of parameters required is 2,477,189, which is for the 2D CNN. The 2D CNNs, which are particularly time-intensive, take 25 s per epoch, whereas the 1D CNNs require between 11 and 22 s per epoch, depending on the kernel size. When combined for all signal modalities, the total processing time amounts to 678 s per epoch.

For a comparative perspective, 3DSleepNet [33], which utilizes multiple 3D and 2D convolutional layers, pseudo-3D convolutions, and partial dot-product attention layers, alongside an LSTM layer, has a training time recorded at 4493 s for 8549 signal segments. This reflects its complex architecture and the computational burden associated with such an extensive epoch count. Similarly, EEGSNet [35], which integrates 5 convolutional blocks and Bi-LSTM layers, operates with 0.6 million parameters, managing to keep computational costs relatively modest through its lower parameter count. In contrast, models such as JK-STGCN [41], which includes adaptive graph learning layers and a jumping knowledge graph convolution module, report a training time of 2621 s for 10 subjects, highlighting a considerable computational demand due to the integration of advanced graph-based techniques. Additionally, CTCNet [40], incorporating multi-scale convolutional neural networks, transformer modules, and capsule networks, also exemplifies a high computational cost due to its multi-component architecture designed for extensive feature extraction and temporal context learning.

4.5. Effect of different feature extraction methods

To scrutinize the impact of our chosen feature extraction methods, we conducted a series of experiments focusing on three different scenarios. In each scenario, we exclusively employed one of the feature extraction methods: raw signals, STFT, or DWT. The results of these experiments are presented in Tables 8 and 9.

Our findings in this phase substantiate the fundamental role of feature extraction methods in sleep stage classification. While each method carries its own unique attributes, it is the synergy among them that gives our algorithm its edge. Raw signals provide a solid grounding in the unprocessed physiological reality, while the STFT offers insights into temporal frequency dynamics, and the DWT dissects signals into time-scale components. This multi-modal fusion creates a feature space that captures the intricacies of sleep dynamics, ultimately contributing to our superior classification performance.

Table 8

Performance Comparison of Feature Extraction Methods in Three Scenarios on Sleep EDF expanded dataset.

Kind of feature used	Overall Accuracy	MF1	Wake	N1	N2	N3	REM
using only RAW signal based features	82.87	75.28	96.44	35.64	92.96	84.92	59.07
using only STFT based features	77.84	69.07	86.61	31.94	89.54	57.34	73.82
using only wavelet based features	76.18	63.23	97.59	22.78	80.73	96.03	36.41

Table 9

Performance Comparison of Feature Extraction Methods in Three Scenarios on ISRUC-SLEEP dataset.

Kind of feature used	Overall Accuracy	MF1	Wake	N1	N2	N3	REM
using only RAW signal based features	85.09	80.28	96.92	52.23	80.59	91.30	79.81
using only STFT based features	83.43	78.65	96.26	53.20	78.02	89.49	75.05
using only wavelet based features	83.07	77.55	97.31	49.13	77.06	89.37	74.86

4.6. Effect of different window and kernel sizes

In this part of our discussion, we examined the impact of varying window and kernel sizes individually. For this analysis, we selected a single window size for STFT feature calculation and a single CNN kernel size in the first layer for wavelet and raw features. This approach allowed us to focus on the isolated effect of different window and kernel sizes on the final results. The results of these experiments are presented in Tables 10 and 11.

Our investigation revealed that the choice of window and kernel sizes significantly influences classification performance. This sensitivity underscores the importance of selecting appropriate sizes that align with the characteristics of the data. Larger windows and kernels enhance frequency discrimination and feature extraction, while smaller sizes provide superior temporal accuracy. The strategic combination of various window and kernel sizes within our algorithm amplifies the potential for improved classification accuracy. This, combined with our multi-modal approach, further emphasizes the robustness and adaptability of our methodology. By employing different sizes in our algorithm, we effectively harness the diversity inherent in the data. This flexibility enables us to capture nuances and subtleties that may be missed when using fixed sizes, ultimately contributing to the superior results observed in our sleep stage classification.

Table 10

Performance when only one window size (STFT calculation) or one Kernel Size (1st Layer of 1D CNN) is used on Sleep EDF expanded dataset.

Kind of feature used	Overall Accuracy	MF1	Wake	N1	N2	N3	REM
using only one of RAW signal based features	79.19	69.88	92.53	31.33	90.30	67.66	59.27
using only one of STFT based features	76.12	66.93	94.31	38.78	76.93	86.51	43.12
using only one of wavelet based features	73.10	59.16	99.15	21.68	73.03	76.19	30.37

Table 11

Performance when only one window size (STFT calculation) or one Kernel Size (1st Layer of 1D CNN) is used ISRUC-SLEEP dataset.

Kind of feature used	Overall Accuracy	MF1	Wake	N1	N2	N3	REM
using only one of RAW signal based features	84.17	79.21	96.21	51.84	78.47	91.43	79.62
using only one of STFT based features	83.16	77.63	96.98	46.41	78.41	90.46	73.90
using only one of wavelet based features	80.55	74.93	94.29	45.83	73.84	89.25	73.14

4.7. Synergy between different signal modalities

In the final part of our discussion, we unveiled the potential of combining features from diverse signal modalities. By individually feeding features derived from various modalities into our network, we showcased the profound impact of multi-modal integration on classification performance. The results of these experiments are presented in [Tables 12 and 13](#).

4.8. Superiority of the proposed model

The proposed model for sleep stage classification demonstrates clear superiority over existing state-of-the-art methods from both theoretical and practical perspectives. This superiority is rooted in the innovative integration of multiple signal modalities and advanced feature extraction techniques. By leveraging Electrooculography (EOG) and two-channel Electroencephalography (EEG) data, our model captures a richer set of physiological signals compared to models that rely on a single modality. This multi-modal approach enhances the model's ability to discern complex sleep patterns, thereby improving classification accuracy. Our model employs a combination of traditional feature extraction methods, such as Short-Time Fourier Transform (STFT) and Discrete Wavelet Transform (DWT), along with neural network-based feature extraction. This hybrid approach capitalizes on the strengths of both traditional signal processing and modern machine learning techniques. Traditional methods provide a robust foundation for analyzing frequency and temporal dynamics, while neural networks automate the extraction of intricate features from raw data. The dual focus on frequency and temporal analysis enables our model to detect subtle changes in sleep stages that other models might miss. Dimensionality reduction through Principal Component Analysis (PCA) plays a crucial role in our methodology. By applying PCA, we effectively reduce the feature space, which not only enhances computational efficiency but also mitigates the risk of overfitting. This ensures that the most informative features are retained, contributing to the overall robustness and reliability of the classifier. In terms of practical performance, our model achieves impressive overall accuracy and macro-F1 scores, particularly

with the XGBoost classifier. The class-specific accuracies for different sleep stages, especially in detecting Wake and N2 stages, highlight the model's proficiency. Furthermore, the model's robustness is validated through evaluations on multiple datasets, including the Sleep-EDF and ISRUC-Sleep databases. The consistent high performance across these datasets underscores the generalizability and applicability of our approach in various settings. The flexibility of our model is evident from its evaluation with a range of classifiers, such as Random Forest, XGBoost, and Long Short-Term Memory (LSTM) networks. This adaptability allows the model to be customized to specific needs and computational resources, making it scalable for diverse applications, from clinical diagnostics to personal health monitoring systems. Compared to State-of-the-Art methods, our model offers a comprehensive analysis by effectively combining traditional signal processing and neural network-based approaches. Many existing methods focus solely on one of these aspects, but our integrated approach results in more accurate and reliable sleep stage classification. Additionally, the multi-modal integration and advanced feature extraction techniques ensure that our model adapts better to variations in data quality and class imbalances, common challenges in sleep stage classification.

In summary, our proposed model excels in both theoretical and practical aspects by integrating a multi-modal approach, combining traditional and neural network-based feature extraction methods, and demonstrating high accuracy and robustness across multiple datasets. These strengths collectively enhance the model's reliability, efficiency, and applicability.

5. Conclusion

This study presents a novel approach to sleep stage classification by integrating multiple signal modalities and advanced feature extraction techniques. Leveraging Electrooculography (EOG) and two-channel Electroencephalography (EEG) data, our model capitalizes on the strengths of both traditional feature extraction methods and neural network-based approaches. The combination of Short-Time Fourier Transform (STFT), Discrete Wavelet Transform (DWT), and raw signal processing with Convolutional Neural Networks (CNNs) allows for a comprehensive analysis of both temporal and spectral features. Our methodology demonstrated superior performance across the Sleep-EDF and ISRUC-Sleep databases, achieving high overall accuracy. Particularly noteworthy is the model's proficiency in detecting Wake and N2 stages, which highlights its robustness and generalizability. Despite the inherent challenges of classifying the N1 stage, our approach showed competitive performance, indicating the potential for further refinement and enhancement. The XGBoost classifier emerged as the top performer, showcasing its effectiveness in handling complex patterns and interactions within the data. The model's flexibility and adaptability were further validated through its evaluation with various classifiers, including Random Forest and Long Short-Term Memory (LSTM) networks. This adaptability ensures that our model can be tailored to specific needs and computational resources, making it suitable for diverse applications from clinical diagnostics to personal health monitoring systems. However, it is important to note that our method involves significant computational costs. The extensive feature extraction stage, which employs multiple neural networks tailored for different signal modalities, results in a high number of parameters and substantial processing time per epoch. This computational demand presents challenges for real-time or resource-constrained applications, highlighting

Table 12

Performance comparison of different signal modalities on Sleep EDF expanded dataset.

Kind of modality used	Overall Accuracy	MF1	Wake	N1	N2	N3	REM
using only EOG signals	73.28	63.66	83.61	19.77	92.14	68.85	50.83
using only EEG Fpz-Cz signals	79.55	71.22	97.06	29.55	86.05	85.91	50.63
using only EEG Pz-Oz signals	77.86	65.65	94.46	16.14	83.58	88.29	63.85

Table 13
Performance comparison of different signal modalities on ISRUC-SLEEP dataset.

Kind of modality used	Overall Accuracy	MF1	Wake	N1	N2	N3	REM
using only EOG signals	81.79	76.35	96.87	49.90	77.38	87.20	65.33
using only EEG C3-A2 signals	78.57	73.17	85.38	48.74	77.31	87.80	73.33
using only EEG F3-A2 signals	81.92	75.79	94.56	42.33	78.28	89.13	76.38

an area for future improvement. In conclusion, this study provides a robust and comprehensive framework for sleep stage classification, offering significant improvements in accuracy and reliability. The innovative integration of multi-modal signal processing and advanced machine learning techniques holds great promise for advancing the field of sleep research and diagnostics. Future research will focus on optimizing the computational efficiency of the model, particularly in improving the detection accuracy of challenging stages like N1, and exploring its applicability to other related domains.

CRedit authorship contribution statement

Mahdi Samaee: Data curation, Formal analysis, Methodology, Project administration, Software, Writing – original draft. **Mehran Yazdi:** Conceptualization, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Daniel Massicotte:** Conceptualization, Project administration, Supervision, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Faust O, et al. A review of automated sleep stage scoring based on physiological signals for the new millennia. *Comput Methods Prog Biomed* 2019;176:81–91.

[2] Rundo JV, Downey III R. *Polysomnography Handbook of clinical neurology* 2019; 160:381–92.

[3] Rechtschaffen A. A manual of standardized terminology, techniques and scoring system for sleep stages of human subjects. *Brain Information Service Brain Research Institute* 1968;10:1–55.

[4] Iber, C., *Ancoli-Israel S*, Chesson A, Quan S(2007) The AASM manual for the scoring of sleep and associated events: Rules, terminology and technical specifications. American Academy of Sleep Medicine.

[5] Delimayanti MK, et al. Classification of brainwaves for sleep stages by high-dimensional FFT features from EEG signals. *Appl Sci* 2020;10(5):1797.

[6] Behzad R, Behzad A. The role of EEG in the diagnosis and Management of Patients with sleep disorders. *Journal of Behavioral and Brain Science* 2021;11(10):257–66.

[7] Smith, S.J., EEG in the diagnosis, classification, and management of patients with epilepsy. *J Neurol Neurosurg Psychiatry*, 2005. 76(suppl 2): p. ii2-ii7.

[8] Ameer A, Saidatul A, Ibrahim Z. Analysis of EEG spectrum bands using power spectral density for pleasure and displeasure state. In: *IOP conference series: Materials science and engineering*. IOP Publishing; 2019.

[9] Basics B. *Understanding sleep*. Bethesda: National Institute of Neurological Disorders and Stroke; 2006.

[10] Lambert I, Peter-Derex L. Spotlight on sleep stage classification based on EEG. *Nature and Science of Sleep* 2023;479–90.

[11] Markun LC, Sampat A. Clinician-focused overview and developments in polysomnography. *Curr Sleep Med Rep* 2020;6:309–21.

[12] Koley B, Dey D. An ensemble system for automatic sleep stage classification using single channel EEG signal. *Comput Biol Med* 2012;42(12):1186–95.

[13] Ronzhina M, et al. Sleep scoring using artificial neural networks. *Sleep Med Rev* 2012;16(3):251–63.

[14] Zoubek L, et al. Feature selection for sleep/wake stages classification using data driven methods. *Biomedical Signal Processing and Control* 2007;2(3):171–9.

[15] Guillot A, Thorey V. RobustSleepNet: transfer learning for automated sleep staging at scale. *IEEE Trans Neural Syst Rehabil Eng* 2021;29:1441–51.

[16] Pascualvaca JMS, et al. Sleep stage classification using advanced intelligent methods. In: *Advances in computational intelligence: 12th international work-conference on artificial neural networks, IWANN 2013, Puerto de la Cruz, Tenerife, Spain, June 12–14, 2013, proceedings, part I 12*. Springer; 2013.

[17] Ghimatgar H, et al. An automatic single-channel EEG-based sleep stage scoring method based on hidden Markov model. *J Neurosci Methods* 2019;324:108320.

[18] Shen H, et al. An automatic sleep stage classification algorithm using improved model based essence features. *Sensors* 2020;20(17):4677.

[19] Pei W, et al. An automatic method using MFCC features for sleep stage classification. *Brain Informatics* 2024;11(1):6.

[20] Supratak A, et al. DeepSleepNet: a model for automatic sleep stage scoring based on raw single-channel EEG. *IEEE Trans Neural Syst Rehabil Eng* 2017;25(11): 1998–2008.

[21] SM, I.N., et al.. *Sleep stage classification based on EEG, EOG, and CNN-GRU deep learning model*. In *2019 IEEE 10th international conference on awareness science and technology (ICAST)*. IEEE 2019:1–7.

[22] Jiang D, et al. Robust sleep stage classification with single-channel EEG signals using multimodal decomposition and HMM-based refinement. *Expert Syst Appl* 2019;121:188–203.

[23] Huang J, et al. Single-channel EEG automatic sleep staging based on transition optimized HMM. *Multimed Tools Appl* 2022;81(30):43063–81.

[24] Eldele E, et al. An attention-based deep learning approach for sleep stage classification with single-channel EEG. *IEEE Trans Neural Syst Rehabil Eng* 2021; 29:809–18.

[25] Jia Z, et al. *Adaptive spatial-temporal graph convolutional networks for sleep stage classification*. In *Ijcai: GraphSleepNet*; 2020.

[26] Olesen, A.N., et al., Automatic sleep stage classification with deep residual networks in a mixed-cohort setting. *Sleep*, 2021. 44(1): p. zsaal61.

[27] You Y, et al. Automatic sleep stage classification: a light and efficient deep neural network model based on time, frequency and fractional Fourier transform domain features. *Artif Intell Med* 2022;127:102279.

[28] Mai X, Yu T. *BootstrapNet: an contrastive learning model for sleep stage scoring based on raw Single-Channel electroencephalogram*. In *2021 2nd international conference on artificial intelligence and computer engineering (ICAICE)*. IEEE 2021:303–8.

[29] Neng W, Lu J, Xu L. CCRRSleepNet: a hybrid relational inductive biases network for automatic sleep stage classification on raw single-channel EEG. *Brain Sci* 2021; 11(4):456.

[30] Zhou D, et al. SingleChannelNet: a model for automatic sleep stage classification with raw single-channel EEG. *Biomedical Signal Processing and Control* 2022;75: 103592.

[31] Lee S, et al. Sleepyco: automatic sleep scoring with feature pyramid and contrastive learning. *Expert Syst Appl* 2024;240:122551.

[32] Supratak A, Guo Y. *TinySleepNet: an efficient deep learning model for sleep stage scoring based on raw single-channel EEG*. In *2020 42nd annual international conference of the IEEE engineering in Medicine & Biology Society (EMBC)*. IEEE 2020:641–4.

[33] Ji X, Li Y, Wen P. 3DSleepNet: a multi-channel bio-signal based sleep stages classification method using deep learning. *IEEE Trans Neural Syst Rehabil Eng* 2023;31:3513–23.

[34] Ji X, et al. MixSleepNet: a multi-type convolution combined sleep stage classification model. *Comput Methods Prog Biomed* 2024;244:107992.

[35] Li C, et al. A deep learning method approach for sleep stage classification with eeg spectrogram. *Int J Environ Res Public Health* 2022;19(10):6322.

[36] Lv X, Li J, Xu Q. A multilevel temporal context network for sleep stage classification. *Comput Intell Neurosci* 2022;2022(1):6104736.

[37] Kong G, et al. EEG-based sleep stage classification via neural architecture search. *IEEE Trans Neural Syst Rehabil Eng* 2023;31:1075–85.

[38] Li R, et al. A developed lstm-ladder-network-based model for sleep stage classification. *IEEE Trans Neural Syst Rehabil Eng* 2023;31:1418–28.

[39] Jia Z, et al. Multi-view spatial-temporal graph convolutional networks with domain generalization for sleep stage classification. *IEEE Trans Neural Syst Rehabil Eng* 2021;29:1977–86.

[40] Zhang W, et al. CTCNet: a CNN transformer capsule network for sleep stage classification. *Measurement* 2024;226:114157.

[41] Ji X, Li Y, Wen P. Jumping knowledge based spatial-temporal graph convolutional networks for automatic sleep stage classification. *IEEE Trans Neural Syst Rehabil Eng* 2022;30:1464–72.

[42] Li W, Gao J. Automatic sleep staging by a hybrid model based on deep 1D-ResNet-SE and LSTM with single-channel raw EEG signals. *PeerJ Computer Science* 2023; 9:e1561.

[43] Xiao Z, et al. RTFN: a robust temporal feature network for time series classification. *Inf Sci* 2021;571:65–86.

[44] Ji Z, et al. A space-embedding strategy for anomaly detection in multivariate time series. *Expert Syst Appl* 2022;206:117892.

[45] Eldele E, et al. ADAST: attentive cross-domain EEG-based sleep staging framework with iterative self-training. *IEEE Transactions on Emerging Topics in Computational Intelligence* 2022;7(1):210–21.

[46] Efe E, Ozsen S. CoSleepNet: automated sleep staging using a hybrid CNN-LSTM network on imbalanced EEG-EOG datasets. *Biomedical Signal Processing and Control* 2023;80:104299.

[47] Zhou D, et al. *LightSleepNet: a lightweight deep model for rapid sleep stage classification with spectrograms*. In *2021 43rd annual international conference of the IEEE engineering in Medicine & Biology Society (EMBC)*. IEEE 2021:43–6.

- [48] Goldberger AL, et al. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *circulation* 2000;101(23): e215–20.
- [49] Kemp B, et al. Analysis of a sleep-dependent neuronal feedback loop: the slow-wave microcontinuity of the EEG. *IEEE Trans Biomed Eng* 2000;47(9):1185–94.
- [50] Khalighi S, et al. ISRUC-sleep: a comprehensive public dataset for sleep researchers. *Comput Methods Prog Biomed* 2016;124:180–92.
- [51] Dong H, et al. Mixed neural network approach for temporal sleep stage classification. *IEEE Trans Neural Syst Rehabil Eng* 2017;26(2):324–33.
- [52] Cohen MX. *Analyzing neural time series data: Theory and practice*. MIT press; 2014.